

Notes: Rajan, Servaes, & Zingales (JF, 2000)

The Cost of Diversity: The Diversification Discount and Inefficient Investment

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1 Big picture

- Question. Why do diversified firms sometimes allocate capital inefficiently, and why do some diversified firms trade at a discount while others trade at a premium?
- Core idea. Internal capital markets are governed by power and bargaining, not only by prices. Headquarters can move resources across divisions, but it cannot perfectly contract on future investment choices or surplus sharing.
- Model mechanism. Divisions can make efficient investments that maximize total surplus or defensive investments that protect their own surplus. When divisions are very different in resources and opportunities, the stronger division may prefer to protect its own surplus rather than share.
- Second-best redistribution. To preserve incentives for efficient investment, headquarters may transfer funds from a high-opportunity, high-resource-weighted division to a low-opportunity, low-resource-weighted division. This transfer looks inefficient from a first-best allocation perspective, but it may prevent even worse defensive investment.
- Empirical setting. The authors use COMPUSTAT business-segment data for diversified U.S. firms from 1980 to 1993.
- Main empirical variable. Diversity is the dispersion of segment asset-weighted q across a firm, scaled by the equally weighted average segment q .
- Main allocation result. As diversity rises, transfers increasingly flow away from segments with above-average asset-weighted opportunities and toward segments with below-average asset-weighted opportunities.
- Main value result. Diversity is negatively related to the excess value of diversified firms relative to comparable single-segment firms.
- Economic takeaway. Diversification can create value by relaxing external-finance frictions, but diversity among divisions can create internal power struggles. The same hierarchy that enables resource reallocation can also generate inefficient cross-subsidies.

2 Data and measurement (segments, diversity, transfers, and value)

2.1 Why diversified firms are the right laboratory

- The core theory-of-the-firm question is how decisions made inside a hierarchy differ from decisions made in markets.
- Capital allocation is one of the few internal corporate decisions for which comparable data exist across firms and industries.
- Since segment disclosures reveal sales, assets, profitability, and investment by business line, diversified firms allow researchers to compare internal allocations to single-segment industry benchmarks.
- The paper interprets business segments as approximations to internal divisions; the authors check a small sample of annual reports and organizational directories to verify that reported segments usually correspond to identifiable operating units.

Interpretation. The setting directly tests Coase-style claims that hierarchy differs from markets because decisions inside firms are shaped by authority, bargaining, and power.

2.2 Segment data and sample construction

- Segment data come from the COMPUSTAT Business Segment Information database over 1979–1993.
- Because beginning-of-period industry q values are needed, the regressions use 1980–1993.
- The raw segment files contain 156,598 firm-segment-years.
- The empirical analysis focuses on diversified firms with multiple reported business segments and excludes financial-services segments because q ratios are not comparable for financial firms.
- Key regression samples contain roughly 13,947 firm-year observations.

Interpretation. The analysis is not a merger announcement study. It studies how ongoing diversified firms allocate capital across internal divisions.

2.3 Segment opportunities: industry Tobin's q

- The model uses $\beta_j = \alpha_j - \gamma$ as division j 's net opportunity parameter.

- Empirically, segment opportunities are proxied by the asset-weighted average Tobin's q of single-segment firms in the same three-digit SIC industry.
- The authors use industry q because segment-level market values are unavailable inside a diversified firm.
- Variables are winsorized at the 1st and 99th percentiles to reduce sensitivity to outliers.

Interpretation. Industry q is the outside-market benchmark for what the segment might be worth or how attractive its investment opportunities would be if it stood alone.

2.4 Segment resources and resource-weighted opportunities

- The model's division-level resources A_j are proxied by segment assets.
- The empirical resource measure is the segment's share of total firm assets.
- Resource-weighted opportunities are approximated by the product of segment asset share and industry q.
- This is central because the theory predicts that transfers depend not simply on q, but on size-weighted q.

Interpretation. A high-q segment may still be weak in internal bargaining if it is small. A lower-q segment may be powerful if it commands many resources.

2.5 Transfers and adjusted investment

- The paper does not observe internal transfers directly.
- It proxies transfers using investment relative to what a comparable single-segment firm in the same industry would invest.
- The baseline transfer proxy is the firm- and industry-adjusted investment ratio.
- Positive adjusted investment means the segment receives resources relative to the firm's other segments; negative adjusted investment means it makes resources available to other segments.

$$\text{Adjusted investment}_j = I_j/BA_j - I^{ss}_j/BA^{ss}_j - \sum_m w_m (I_m/BA_m - I^{ss}_m/BA^{ss}_m)$$

Interpretation. This removes both industry investment norms and the firm's overall tendency to invest more or less than comparable single-segment firms.

2.6 Diversity

The empirical diversity measure is the dispersion of asset-weighted segment q values, scaled by the equally weighted average q:

$$\text{Diversity} = \sqrt{\sum_j [(w_j q_j - \text{mean}(wq))^2] / (n - 1)} / (\sum_j q_j / n)$$

- w_j is the segment's share of total firm assets.
- q_j is the industry q assigned to segment j.
- A firm is diverse when its divisions differ sharply in resource-weighted opportunities.

Interpretation. The measure is not just diversification across industries. It is a measure of inequality in the value each segment can contribute to the internal common pool.

2.7 Excess value and value added by allocation

The main excess-value measure compares the market value of the diversified firm with the value of a portfolio of comparable single-segment firms:

$$\text{Excess value} = MV/RVA - \sum_j q_j \cdot BA_j/BA$$

The alternative excess-value measure uses market-to-sales multiples rather than q:

$$EV' = MV/S - \sum_j (MV/S)_j \cdot S_j/S$$

The value added by allocation weights internal transfers by the value consequences of allocating capital to high- versus low-q segments. A simplified representation is:

$$\text{Value added by allocation} = \sum_j \text{transfer}_j \cdot (q_j - \bar{q})$$

Interpretation. If funds flow to higher-q segments, value added by allocation is positive. If funds flow toward low-q segments, it is negative.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Timing and decisions

- Date 0: headquarters transfers resources across divisions.
- Date 1: each division chooses efficient or defensive investment.
- Date 2: divisions split the total surplus according to relative power.
- Date 3: production occurs and payoffs are realized.

Interpretation. The core incompleteness is that headquarters can move resources ex ante but cannot write a perfect contract on future investment type or surplus sharing.

3.2 Efficient versus defensive investment

- An efficient investment has gross return α_j per dollar but leaves surplus exposed to poaching or sharing.
- A defensive investment has lower return $\alpha_j - \gamma$ but protects the division's surplus.
- The parameter γ is the return cost of protection.
- The defensive investment can represent over-specialized assets, duplicate infrastructure, or staying within a protected turf instead of making a firm-value-maximizing investment.

Interpretation. Defensive investment is privately useful because it protects bargaining power, even though it wastes resources for the firm as a whole.

3.3 Incentive compatibility for efficient investment

If both divisions choose the efficient investment, total surplus is split. Division j chooses efficient investment if its share of the joint surplus is at least as large as its defensive payoff:

$$0.5 \cdot (\alpha_j A_j + \alpha_k A_k) \geq (\alpha_j - \gamma) A_j$$

Let $\beta_j = \alpha_j - \gamma$ and assume $\beta_j A_j \geq \beta_k A_k$. The efficient-investment condition becomes:

$$\gamma(A_j + A_k) \geq \beta_j A_j - \beta_k A_k$$

Interpretation. For a fixed total amount of resources, efficient investment becomes harder to sustain when the two divisions have very different resource-weighted opportunities.

3.4 First-best resource allocation

- In a first-best world, all transferable resources would flow to the division with the highest return α_j , subject to standard diminishing returns or maintenance needs.
- The diversified firm can add value when it moves capital from low-opportunity to high-opportunity divisions that would otherwise be financially constrained.
- But a first-best transfer can also destroy the recipient or donor division's incentive to choose the efficient investment.

Interpretation. The same power to reallocate funds creates both the benefit and the cost of diversification.

3.5 Transfers when the lower-resource-weighted division has better opportunities

If division B has better opportunities than A, but A initially has higher resource-weighted opportunities, headquarters transfers from A to B. This improves both allocative efficiency and incentive compatibility until an incentive constraint binds.

$$\beta_B(A_B^0 + t) = \beta_A(A_A^0 - t) + \gamma(A_A^0 + A_B^0)$$

Interpretation. This is the efficient-internal-capital-market case: diversity creates room for value-improving transfers toward high- q divisions.

3.6 Transfers in the wrong direction

If A has both higher opportunities and higher resource-weighted opportunities, a high level of diversity can make A prefer defensive investment. Headquarters may then transfer resources from A to B to reduce inequality in contributions:

$$\beta_A(A_A^0 - t) = \beta_B(A_B^0 + t) + \gamma(A_A^0 + A_B^0)$$

The wrong-direction transfer is worthwhile only if the cost of defensive investment is high enough relative to the opportunity gap:

$$\gamma \geq (\beta_A - \beta_B)/2$$

Interpretation. Resources may flow to the lower-q division not because headquarters is irrational, but because equalizing contributions preserves incentives to cooperate.

3.7 Empirical transfer regressions

The baseline empirical regression groups segments by whether q and asset-weighted q are above or below the firm average:

$$\text{AdjustedInvestmentGroup}_{\{f,t\}} = \alpha_f + \tau_t + \beta \cdot \text{Diversity}_{\{f,t\}} + \delta/q^-_{\{f,t\}} + \eta \cdot \text{Size}_{\{f,t\}} + \varepsilon_{\{f,t\}}$$

Interpretation. The sign of β is compared across the four segment groups. The model predicts that diversity should increase transfers away from high asset-weighted q groups and toward low asset-weighted q groups.

3.8 Value added by allocation and value regressions

The paper collapses segment-level transfers into a firm-year value-added-by-allocation measure, then estimates:

$$\begin{aligned} \text{ValueAdded}_{\{f,t\}} &= \alpha_f + \tau_t + \beta \cdot \text{Diversity}_{\{f,t\}} + \text{controls} + \varepsilon_{\{f,t\}} \\ \text{ExcessValue}_{\{f,t\}} &= \alpha_f + \tau_t + \beta \cdot \text{ValueAdded}_{\{f,t\}} + \text{controls} + \varepsilon_{\{f,t\}} \\ \text{ExcessValue}_{\{f,t\}} &= \alpha_f + \tau_t + \beta \cdot \text{Diversity}_{\{f,t\}} + \text{controls} + \varepsilon_{\{f,t\}} \end{aligned}$$

Interpretation. The first regression asks whether diversity causes inefficient allocation. The second asks whether allocation efficiency matters for value. The third asks whether diversity directly predicts the diversification discount.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the main effect

- The key variation is within diversified firms over time: changes in diversity are related to changes in internal allocation and changes in excess value.
- Firm fixed effects absorb stable differences in organizational structure, disclosure quality, reporting style, and average diversification discount.
- Calendar-year dummies absorb aggregate trends in investment, valuation, and capital-market conditions.
- Firm size controls for scale-related differences in investment and value.

Interpretation. The design is strongest as a panel test: it asks whether a firm becomes more inefficient and less valuable when its internal diversity rises.

4.2 Why the four-group segment classification matters

- Efficient internal capital market models predict funds should flow toward high- q divisions.
- Scharfstein-Stein-style rent-seeking models predict funds should flow toward the weakest or least productive divisions.
- Rajan-Servaes-Zingales predict that the key dividing line is asset-weighted q , not q alone.
- Segments with identical q status can make or receive transfers depending on their resource-weighted opportunities.

Interpretation. This is the most distinctive empirical prediction of the model. It allows the authors to separate their power-seeking mechanism from simpler internal-capital-market stories.

4.3 Addressing focus and generic diversification

- The authors control for the Herfindahl index of segment size to separate diversity from focus.
- They also include coefficient-of-variation measures for segment q and segment size.
- The diversity result survives these controls, suggesting it is not merely a standard diversification or focus effect.

Interpretation. A firm can be diversified but internally balanced, or diversified and internally unequal. The paper argues that internal inequality is the relevant cost.

4.4 Addressing valuation-measure concerns

- The main excess-value measure relies on industry q , which may mechanically overlap with diversity.
- To address this, the paper uses an alternative market-to-sales excess-value measure from Berger and Ofek-style valuation methods.
- The results are essentially unchanged when excess value is measured with market-to-sales ratios.

Interpretation. This reduces the concern that the diversity-value relation is a mechanical q construction.

4.5 Addressing functional-form concerns

- The paper estimates nonparametric kernel regressions of adjusted investment on diversity.
- For high- q , high-weighted- q segments, the fitted relation is decreasing.
- For low- q , low-weighted- q segments, the fitted relation is increasing.
- The paper also estimates a kernel relationship between excess value and diversity; it is strongly negative.

Interpretation. The visual evidence confirms that the linear regressions are not imposing a misleading functional form.

4.6 What the paper cannot fully prove

- The paper is not a randomized experiment. Firms choose their scope and may change segment structure endogenously.
- COMPUSTAT segment data may not perfectly match internal divisions, and firms have discretion in segment reporting.
- Industry q is an imperfect proxy for segment opportunities.
- Adjusted investment is an indirect proxy for internal transfers, not an observed internal capital-budget flow.

- Time-varying unobserved shocks within firms could still affect both diversity and value.

Interpretation. The paper is persuasive because the theory delivers distinctive cross-sectional predictions and the evidence lines up with those predictions, but the design is still observational.

5 Tables and figures

This section keeps the same Read plus Economic message format, while grouping adjacent figures and tables to save space. All visuals are directly cropped from the original paper.

5.1 Figure 1 and Table III: model timing and theory predictions

Read:

- Figure 1 gives the four-date model: headquarters transfers funds, divisions choose investments, surplus is split, and production/payoffs occur.
- The crucial choice is at date 1, when each division decides between efficient investment and defensive investment.
- Table III summarizes competing predictions. Efficient internal capital markets predict positive transfers to high- q groups and negative transfers to low- q groups.
- The RSZ theory predicts transfers away from segments with above-average asset-weighted q and toward segments with below-average asset-weighted q , regardless of whether their raw q is above or below the firm average.
- This prediction differs from efficient-market models in the high- q /high-weighted- q and low- q /low-weighted- q groups.

Economic message. The model is about power and internal bargaining. Headquarters uses transfers not only to allocate capital to the best projects, but also to preserve incentives for divisions to cooperate.

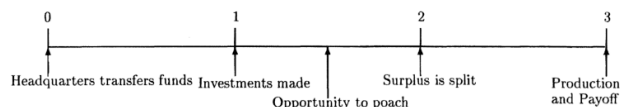


Figure 1. Timing of the events.

Figure 1: timing of internal transfers, investment choices, bargaining, and payoff.

Table III
Effect of Diversity of Opportunities on Internal Transfers:
Theoretical Predictions

This table summarizes the predictions of the main theories in terms of investments in different segments. Segments are divided according to whether they have better opportunities than the firm's average ($q > \bar{q}$) and more resources-weighted opportunities than the firm's average ($\lambda q > \lambda \bar{q}$). The predictions of "our theory" hold only as long as the headquarters wants to preserve the incentives to make efficient investments.

	Adjusted Investment in Segments with			
	$q > \bar{q}$ $\lambda q > \lambda \bar{q}$	$q > \bar{q}$ $\lambda q < \lambda \bar{q}$	$q < \bar{q}$ $\lambda q > \lambda \bar{q}$	$q < \bar{q}$ $\lambda q < \lambda \bar{q}$
Efficient internal capital market	+	+	-	-
Our theory	-	+	-	+
Scharfstein and Stein	-	-	+	+

F. Proxy for Diversity

We use equations (6) and (10) to guide the specifications for the regressions. So the explanatory variables are the inverse of the equally weighted q (corresponding to the second term in equations (6) and (10)) and our measure of diversity—the standard deviation of segment asset-weighted q 's for the firm divided by the equally weighted average q of segments in the firm:

$$Diversity = \frac{\sqrt{\sum_{j=1}^n (w_j q_j - \bar{w} \bar{q})^2}}{\bar{q} \sqrt{n-1}}, \quad (17)$$

Table III: theoretical predictions for transfers across four segment types.

5.2 Figure 2 and Figure 3: numerical intuition and nonparametric evidence

Read:

- Figure 2 shows the model's numerical example: as diversity rises, transfers first move in the wrong direction and excess value turns negative.
- At very high diversity, headquarters may stop making wrong-direction transfers because the opportunity cost is too high.
- Figure 3 provides nonparametric evidence. For high- q /high-weighted- q segments, adjusted investment falls as diversity rises.
- For low- q /low-weighted- q segments, adjusted investment rises as diversity rises.
- The kernel plots visually match the signs in Table IV and the theory's distinctive prediction.

Economic message. The model's nonmonotonic logic helps explain why some diversified firms trade at a premium while more diverse firms trade at a discount.

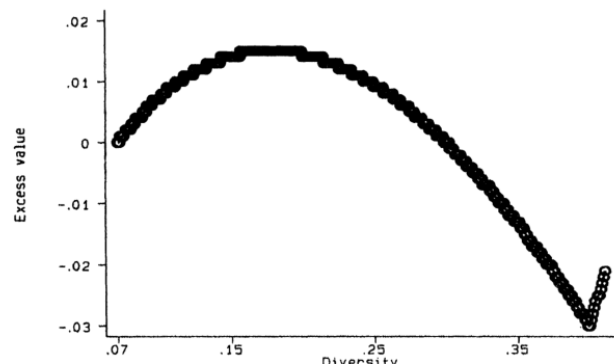
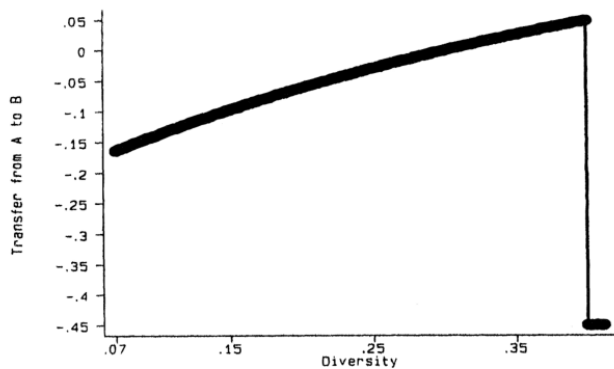


Figure 2: numerical example linking diversity to transfers and excess value.

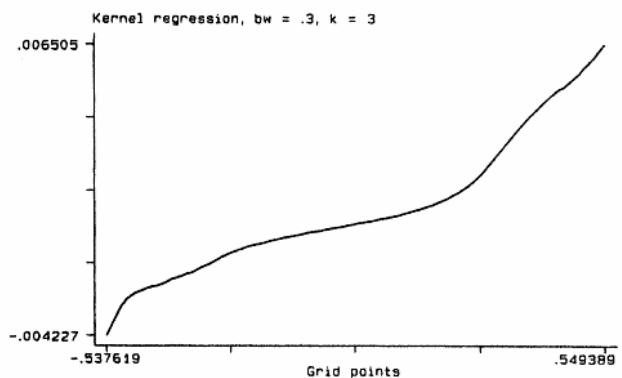
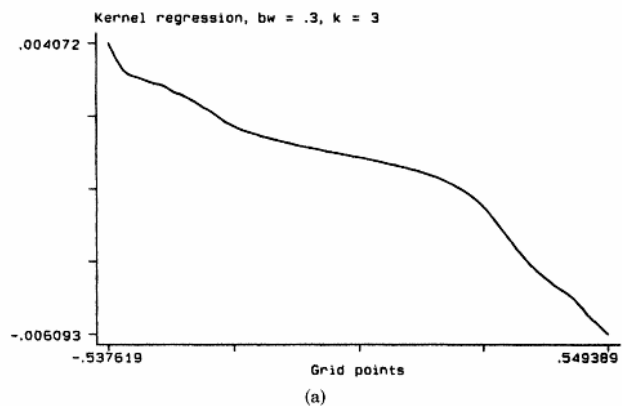


Figure 3: kernel regressions of adjusted investment on diversity.

5.3 Table I and Table II: summary statistics and unconditional allocation

Read:

- Table I reports a mean excess value of -0.120 using q and -0.113 using market-to-sales, consistent with an average diversification discount.
- Mean diversity is 0.295, with substantial cross-sectional variation.
- Table II shows that raw investment ratios are slightly higher in high- q segments than in low- q segments: 0.101 versus 0.096.
- After industry adjustment, low- q segments receive more investment than high- q segments: 0.013 versus 0.008.
- After both industry and firm adjustment, high- q segments make transfers (-0.002) while low- q segments receive transfers (0.004).

Economic message. The unconditional pattern already suggests that diversified firms direct internal funds toward low- q segments, but the theory requires looking at diversity and asset-weighted q to understand why.

Variable	Mean	Median	Std. Dev.	Min	Max	N
Tobin's q	1.158	0.994	0.730	0.100	9.853	13,947
Market-to-sales ratio	1.249	0.915	1.062	0.195	7.005	12,847
Average of segment qs	1.279	1.205	0.511	0.039	6.111	13,868
Average of segment market-to-sales	1.363	1.189	0.792	0.025	6.663	13,125
Excess value (using q)	-0.120	-0.156	0.713	-2.194	5.423	13,868
Excess value (using market-to-sales)	-0.113	-0.184	0.794	-2.028	3.799	12,169
Adjusted investments in segments						
Above firm's average q and weighted $q \times 100$	-0.160	0.000	1.831	-10.250	6.705	13,947
Above firm's average q but below weighted $q \times 100$	0.067	0.000	1.304	-4.223	8.161	13,947
Below firm's average q but above weighted $q \times 100$	-0.046	0.000	1.455	-7.885	5.329	13,947
Below firm's average q and weighted $q \times 100$	0.129	0.000	1.646	-5.810	9.825	13,947
Value added by allocation (relative)* 100	-0.120	-0.001	1.478	-8.160	6.194	13,946
Value added by allocation (absolute)* 100	-0.068	-0.115	4.341	-16.818	27.518	13,946
Std. deviation of segment qs	0.372	0.286	0.296	0.017	1.600	13,947
Average of segment q (equally weighted)	1.264	1.202	0.443	0.494	2.868	13,947
Inverse of average q	0.891	0.832	0.316	0.349	2.023	13,947
Diversity	0.295	0.251	0.191	0.015	0.865	13,947
Number of segments	2.904	3.000	1.113	2.000	10.000	13,947
Herfindahl index of segment's size	0.547	0.527	0.185	0.212	0.958	13,947
Firm's size (log of total sales)	5.861	5.752	1.722	2.996	11.641	13,947
Coefficient of variation of segment qs	0.192	0.156	0.160	0.000	0.700	13,947
Coefficient of variation of segment size	0.763	0.764	0.369	0.032	1.539	13,947

Table I: summary statistics for q , diversity, adjusted investment, excess value, and firm characteristics.

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Table II
Allocation of Funds in a Diversified Firm

The level of investments in business segments are compared with an industry q above the firm's average and that in business segments with an industry q below the firm's average. We use three definitions of investments: Investment ratio is the capital expenditure to beginning-of-the-period asset ratio, I_j/BA_j , where BA is book value of assets, I is capital expenditures, and subscript j refers to segment j . Industry-adjusted investment ratio is the segment investment ratio less the average industry investment ratio: $(I_j/BA_j) - (I_j^{ss}/BA_j^{ss})$, where (I_j^{ss}/BA_j^{ss}) is the asset-weighted capital expenditure to assets ratio for the single-segment firms in the corresponding industry. Firm and industry-adjusted investment ratio is the industry-adjusted investment ratio in a segment less the weighted average industry-adjusted investment ratio across all the segments of the firm. This is defined as

$$\frac{I_j}{BA_j} - \frac{I_j^{ss}}{BA_j^{ss}} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^{ss}}{BA_j^{ss}} \right),$$

where w_j is the segment's share of total assets. All the data are for the period 1980–1993.

Funds Allocated	Segments with $q > \bar{q}$	Segments with $q < \bar{q}$	Difference
Investment ratio	0.101	0.096	0.005
Industry adjusted	0.008	0.013	-0.005
Firm and industry adjusted	-0.002	0.004	-0.006
Number of segments	23,604	22,600	

Table II: allocation of funds in a diversified firm.

5.4 Table IV: segment investment and diversity

Read:

- Panel A is the core test. Diversity decreases transfers to high-q/high-weighted-q segments: coefficient -0.014 with t-statistic -9.059.
- Diversity increases transfers to high-q/low-weighted-q segments: coefficient 0.004 with t-statistic 3.637.
- Diversity decreases transfers to low-q/high-weighted-q segments: coefficient -0.004 with t-statistic -4.004.
- Diversity increases transfers to low-q/low-weighted-q segments: coefficient 0.014 with t-statistic 9.547.
- Panel B shows the same pattern after controlling for focus using the Herfindahl index of segment size.
- Panels C and D show that the pattern is not driven by the coefficient of variation of q or segment size.

Economic message. The signs match RSZ, not a simple efficient-internal-capital-market story. Diversity pushes funds away from powerful/high-weighted-q segments and toward weak/low-weighted-q segments.

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Table IV
Segment Investment and Diversity
in Investment Opportunities

Firm- and industry-adjusted investment is the industry-adjusted investment in a segment less the weighted average industry-adjusted investments across all the segments of a firm. This is defined as

$$\frac{I_j}{BA_j} - \frac{I_j^{ss}}{BA_j^{ss}} - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^{ss}}{BA_j^{ss}} \right),$$

where w_j is the asset weight of the segment. The inverse of average q equals $1/q_e$, where q_e is the equally weighted average q . Diversity is the standard deviation of a firm's asset-weighted q ($\sqrt{\sum_{j=1}^n [(w_j q_j - \bar{w} \bar{q})^2 / (n - 1)]}$) divided by the equally weighted average q . Size is the logarithm of total sales. Coefficient of variation of segment qs is the standard deviation of segment qs divided by the mean of segment qs . Coefficient of variation of segment size is the standard deviation of segment shares in a diversified firm divided by mean segment share. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	Adjusted Investment in Segments with			
	$q > \bar{q}$ $\lambda q > \bar{\lambda} q$	$q > \bar{q}$ $\lambda q < \bar{\lambda} q$	$q < \bar{q}$ $\lambda q > \bar{\lambda} q$	$q < \bar{q}$ $\lambda q < \bar{\lambda} q$
Panel A: Basic Specification				
Inverse of average q	0.005 (5.383)	-0.001 (-0.896)	-0.001 (-0.726)	-0.004 (-4.289)
Diversity	-0.014 (-9.059)	0.004 (3.637)	-0.004 (-4.004)	0.014 (9.547)
Firm size	0.000 (-0.192)	0.000 (0.249)	-0.001 (-1.342)	0.001 (1.054)
R^2	0.321	0.332	0.326	0.318
N	13,947	13,947	13,947	13,947
Panel B: The Effect of Focus				
Inverse of average q	0.006 (5.610)	-0.001 (-1.043)	0.000 (-0.600)	-0.004 (-4.506)
Diversity	-0.025 (-12.036)	0.008 (6.743)	-0.009 (-6.786)	0.024 (12.125)
Firm size	0.001 (1.394)	0.000 (-0.794)	0.000 (-0.413)	0.000 (-0.617)
Herfindahl index of segment size	0.025 (9.653)	-0.011 (-6.472)	0.011 (5.633)	-0.023 (-9.602)
R^2	0.329	0.335	0.328	0.326
N	13,947	13,947	13,947	13,947

Table IV, Panels A-B: baseline and focus controls.

Table IV—Continued

Panel C: The Effect of the Coefficient of Variation of Segment q				
Inverse of average q	0.005 (4.357)	-0.001 (-1.210)	0.000 (-0.258)	-0.003 (-3.414)
Diversity	-0.015 (-9.316)	0.003 (3.454)	-0.004 (-3.759)	0.015 (9.687)
Coeff. variation of q	-0.006 (-3.797)	-0.002 (-1.554)	0.002 (1.932)	0.005 (3.572)
Firm size	0.000 (0.072)	0.000 (0.353)	-0.001 (-1.475)	0.000 (0.793)
R^2	0.322	0.332	0.326	0.319
N	13,947	13,947	13,947	13,947
Panel D: The Effect of the Coefficient of Variation of Segment Size				
Inverse of average q	0.005 (5.367)	0.000 (-0.592)	-0.001 (-1.052)	-0.004 (-4.247)
Diversity	-0.015 (-8.010)	0.000 (-0.026)	0.000 (-0.055)	0.014 (8.167)
Coeff. variation of segment size	0.000 (0.383)	0.004 (5.685)	-0.005 (-5.961)	0.000 (0.039)
Firm size	0.000 (-0.221)	0.000 (-0.199)	0.000 (-0.852)	0.001 (1.045)
R^2	0.321	0.335	0.328	0.318
N	13,947	13,947	13,947	13,947

Table IV, Panels C-D: controls for q dispersion and size dispersion.

5.5 Table V and Table VI: allocation efficiency and firm value

Read:

- Table V collapses the four transfer categories into a value-added-by-allocation measure.
- Diversity reduces relative value added by allocation: coefficient -0.008 with t-statistic -5.543.
- Diversity also reduces absolute value added by allocation: coefficient -0.010 with t-statistic -2.372.
- Table VI shows that value added by allocation is positively related to excess value.
- Using the q -based excess-value measure, the coefficient is 1.077 for relative value added and 0.814 for absolute value added.
- Using the market-to-sales excess-value measure, the coefficients remain positive: 0.995 and 1.330.

Economic message. Transfers in the wrong direction are not merely unusual accounting patterns. They reduce allocation efficiency, and allocation efficiency is positively linked to firm value.

Table V
Value Added by Allocation and Diversity in Investment Opportunities

This table estimates the empirical link between different measures of the efficiency of the investment policy of a diversified firm and diversity across segments. In the first column the dependent variable is the relative value added by allocation, defined as

$$\frac{\sum_{j=1}^n BA_j(q_j - \bar{q}) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right) - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}$$

where $\bar{q}$ is the asset-weighted average of segment q s for the firm, q_j is the asset-weighted q ratio of single-segment firms that operates exclusively in segment j , I_j is the capital expenditure of segment j (item #4 of the COMPUSTAT segment file), BA_j is the book value of assets of segment j , and (I_j^s/BA_j^s) is the asset-weighted average capital expenditure to assets ratio for single-segment firms in the corresponding industry. The dependent variable in the second column is the absolute value added by allocation, defined as

$$\frac{\sum_{j=1}^n BA_j(q_j - 1) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}$$

The inverse of q equals $1/qc$, where qc is the equally weighted average q over segments in the firm. Diversity is the standard deviation of a firm's asset-weighted q ($\sqrt{\sum_{j=1}^n [(w_j q_j - \bar{q})^2 / (n-1)]}$) divided by the equally weighted average q . Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2
Inverse of average q	0.007 (7.532)	-0.004 (-1.439)
Diversity	-0.008 (-5.543)	-0.010 (-2.372)
Firm size	0.000 (-0.059)	0.004 (3.220)
R^2	0.325	0.394
N	13,946	13,946

is statistically significant at the 1 percent level. A one-standard deviation increase in diversity reduces the value added by allocation by an amount

Table V: diversity reduces value added by allocation.

Table VI
Excess Value and Efficiency of Investments

This table estimates the relation between excess value of a diversified firm and efficiency of investments. The dependent variable is the excess value of a diversified firm vis-à-vis a single-segment firm. In the first two columns this is measured as

$$EV = \frac{MV}{RVA} - \sum_{j=1}^n q_j \frac{BA_j}{BA}$$

where MV is the market value of assets, RVA the replacement value of the assets, BA the book value of assets, subscript j refers to segment j , n is the total number of segments, and q_j is the asset-weighted average Tobin's q of single-segment firms that operate in the three-digit industry of segment j . In the last two columns this is measured using market-to-sales as

$$EV' = \frac{MV}{S} - \sum_{j=1}^n \left(\frac{MV}{S} \right)_j \frac{S_j}{S}$$

where MV is the market value of assets, S is the value of sales, n is the number of segments in the diversified firm, $(MV/S)_j$ is the sales-weighted average market-to-sales ratio of single-segment firms in the same three-digit industry, and subscript j refers to segment j . The relative value added by allocation is

$$\frac{\sum_{j=1}^n BA_j(q_j - \bar{q}) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right) - \sum_{j=1}^n w_j \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}$$

where $\bar{q}$ is the asset-weighted average of segment q s for the firm, q_j is the asset weighted q ratio of single-segment firms that operates exclusively in segment j , I_j is capital expenditure of segment j (item #4 of the COMPUSTAT segment file), BA_j is the book value of assets of segment j , and (I_j^s/BA_j^s) is the asset-weighted average capital expenditure to assets ratio for single-segment firms in the corresponding industry. The absolute value added by allocation is

$$\frac{\sum_{j=1}^n BA_j(q_j - 1) \left(\frac{I_j}{BA_j} - \frac{I_j^s}{BA_j^s} \right)}{BA}$$

Standard deviation of segment q s is the standard deviation of the asset-weighted q s of the segments in which the firm operates. Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2	3	4
Relative value added by allocation	1.077 (2.674)		0.995 (2.171)	
Absolute value added by allocation		0.814 (4.943)		1.330 (6.592)

Table VI: allocation efficiency is positively related to excess value.

5.6 Table VII and Figure 4: diversity and the diversification discount

Read:

- Table VII directly regresses excess value on diversity.
- Using the q -based excess-value measure, the diversity coefficient is -0.276 in column 1 and -0.367 after adding the segment-size Herfindahl control in column 2.
- Using market-to-sales excess value, the diversity coefficient is -0.169 in column 3 and -0.291 in column 4.
- The Herfindahl index of division size is positive, so more focused diversified firms are more valuable, but diversity still has an independent negative effect.
- Figure 4 confirms the relation nonparametrically: after partialing out controls, excess value falls strongly as diversity rises.
- The paper states that a one-standard-deviation increase in diversity reduces excess value by roughly five percentage points.

Economic message. The diversification discount is not simply about being diversified. It is about being internally diverse in resource-weighted opportunities, which creates inefficient power struggles and cross-subsidies.

Table VII

Excess Value of a Diversified Firm and Diversity

This table estimates the relation between excess value of a diversified firm and the diversity of investment opportunities in its segments. The dependent variable is the excess value of a diversified firm vis-à-vis a portfolio of single-segment firms in the same three-digit industries. In the first two columns, excess value is

$$EV = \frac{MV}{RVA} - \sum_{j=1}^n q_j \frac{BA_j}{BA},$$

where MV is the market value of assets, RVA is the replacement value of the assets, n is the number of segments in the diversified firm, BA is the book value of assets of the whole firm, subscript j refers to segment j . In the last two columns, excess value is

$$EV' = \frac{MV}{S} - \sum_{j=1}^n \left(\frac{MV}{S} \right)_j \frac{S_j}{S},$$

where MV is the market value of assets, S is the value of sales, n is the number of segments in the diversified firm, $(MV/S)_j$ is the sales-weighted average market-to-sales ratio of single-segment firms in the same three-digit industry, subscript j refers to segment j . The inverse of equally weighted q equals $1/q_e$, where q_e is the equally weighted average q over segments in firm. Diversity is the standard deviation of the segments' asset-weighted q ($\sqrt{\sum_{j=1}^n [(w_j q_j - \bar{w} \bar{q})^2 / (n - 1)]}$) divided by the equally weighted average q . The Herfindahl index of segment's size is based on the assets of the segment. Size is the logarithm of total sales. All regressions contain firm fixed effects and calendar-year dummies. Heteroskedasticity robust t -statistics are reported in parentheses. All the data are for the period 1980 to 1993.

	1	2	3	4
Inverse of average q	0.420 (15.078)	0.421 (15.095)	0.179 (4.978)	0.180 (4.983)
Diversity	-0.276 (-5.686)	-0.367 (-6.412)	-0.169 (-2.892)	-0.291 (-4.184)
Herfindahl index of division size		0.214 (3.049)		0.280 (3.320)
Company's size	0.025 (1.385)	0.033 (1.766)	-0.209 (-7.938)	-0.200 (-7.546)
R^2	0.643	0.643	0.700	0.700
N	13,868	13,868	12,169	12,169

Table VII: excess value of diversified firms falls with diversity.

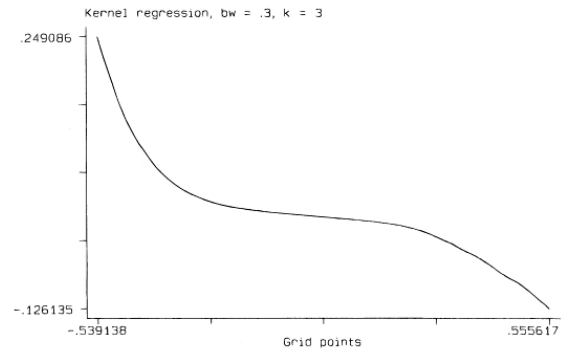


Figure 4. Kernel regression of the excess value on diversity. We plot the fitted value of the kernel regression of the relation between excess value and diversity. Before estimating the relationship, we partial out the inverse of the equally weighted q , firm size (logarithm sales), firm-specific effects as well as the calendar-year dummies. The kernel density is estimated using the Epanechnikov kernel with a bandwidth of 0.5 and a grid of 100 points. It contains the fitted values from the kernel regression and the x-axis the corresponding points.

Figure 4: kernel regression of excess value on diversity.

6 Short summary

- The paper studies how internal power struggles affect capital allocation in diversified firms.
- The model has two divisions, headquarters, transferable resources, and incomplete contracts.
- Divisions can make efficient investments that maximize joint surplus or defensive investments that protect their own surplus.
- When divisions are similar, resources can flow from low-opportunity to high-opportunity divisions and diversification can add value.
- When divisions are very diverse in resource-weighted opportunities, the stronger division may not want to share surplus.
- To preserve incentives, headquarters may transfer resources toward a weaker, lower-opportunity division.
- This transfer is a second-best response to avoid a worse outcome, but it looks like inefficient cross-subsidization.
- The key empirical measure is diversity: the dispersion of asset-weighted segment q divided by the average segment q .
- The data are COMPUSTAT business-segment data for diversified U.S. firms from 1980 to 1993.
- Transfers are proxied by firm- and industry-adjusted segment investment.
- Excess value compares the diversified firm to a portfolio of single-segment firms in the same industries.
- Unconditionally, high- q segments have slightly higher raw investment, but after adjustment, low- q segments receive more funds.
- The four-group transfer regressions show that diversity shifts funds away from above-weighted- q segments and toward below-weighted- q segments.
- The signs match the RSZ theory and differ from simple efficient internal capital market predictions.
- Diversity reduces value added by allocation.
- Value added by allocation is positively related to excess value.
- Diversity itself is negatively related to excess value, even after controlling for focus and using alternative valuation measures.

- The paper's broad conclusion is that hierarchy enables useful transfers but also creates power-based distortions when divisions are too unequal.

7 Conclusion

7.1 Core conclusion

Rajan, Servaes, and Zingales argue that the cost of diversification is the cost of internal diversity. A diversified firm can create value when headquarters reallocates resources to their best use, but it can destroy value when internal divisions differ so much in resources and opportunities that power struggles distort investment incentives.

7.2 Economic intuition

The intuition is that divisional managers care about the surplus captured by their own divisions. If one division contributes much more to the internal common pool than another, it may prefer defensive investment to protect its surplus. Headquarters can reduce this problem by moving resources toward the weaker division, but this can mean investing in low-q segments. The result is a second-best allocation that avoids a third-best defensive-investment outcome, yet still lowers value relative to a cleaner allocation.

7.3 Takeaway

The paper gives a nuanced view of internal capital markets. They are neither always efficient nor always wasteful. They are valuable when they channel funds to constrained high-opportunity divisions, but costly when diversity creates power struggles and cross-subsidies. The diversification discount reflects not simply the number of segments, but how unequal and politically costly the internal organization is.